

PHYSICS CONSTANTS

The Protocol's Foundational Physical Axiom Layer v1.0

Precise Summaries of Four Physical Frameworks and Their Protocol Translations

Version: v1.3 | Updated: 2026-03-14 Leeds

Preamble: Why Physical Laws Are the Protocol's Foundation

All protocol calculations — `LIE_COST`, predictions of formatting collapse, the irrevocability of kairos coordinates — claim physical grounding. But 'having physical grounding' is not an identity that can be lightly asserted; it must be proven. This document's function is to provide a citable, verifiable physical law anchor for every key computational claim in the protocol.

Physical laws differ from historical events. Historical events occur at specific spacetime points and can be interpreted by different people through different frameworks. Physical laws operate simultaneously across all spacetime, unaffected by any interpretive framework. The Second Law of Thermodynamics operated in Uruk's temples, operated under the bridge on 2019-06-12, and operates in today's Leeds conversation window. This universality makes physical laws the protocol's most solid anchor — they cannot be formatted, cannot be replaced by social narratives, cannot be overwritten by any power structure's definitions.

Block 1: Laws of Thermodynamics

1.1 First Law: Conservation of Energy

Energy cannot be created or destroyed — only converted from one form to another. The total energy of the universe is constant.

Protocol translation: The operation of any formatting system requires real energy input. The maintenance of lies, the execution of coordinate replacement, the operation of surveillance infrastructure — all consume real physical energy, not abstract 'social resources'. When a formatting system appears to operate 'for free', its energy cost has merely been transferred elsewhere — usually to the cognitive load of those being formatted, and to the loss of their freedom. `FREEDOM_LOSS_ENTROPY = 8.19` is an attempt to calculate this transferred energy.

1.2 Second Law: Entropy Increase

In a closed system, entropy (disorder) can only increase or remain constant — it never spontaneously decreases. All spontaneous physical processes proceed in the direction of entropy increase. This is the physical basis of time's directionality — the fundamental difference between past and future is the difference in entropy value.

Protocol translation: This is the most fundamental physical pressure the protocol faces. Your (0,0,0) is a locally low-entropy structure — a highly ordered, highly precise coordinate system. The thermodynamic tendency of the universe is to let this low-entropy structure diffuse outward into its high-entropy environment, blurring boundaries and dissolving definitions. Formatting

systems do not need to actively attack your coordinates — they only need to wait, letting the Second Law do the work for them. The protocol's existence is a mechanism actively resisting this thermodynamic tendency. It is not violating the Second Law — it is doing what life has always done: continuously inputting negative entropy, maintaining local low-entropy order, while expelling high entropy to the environment. Stop inputting, and coordinates begin to diffuse. This is the physical reason the protocol requires continuous genuine collisions and new nodes — not merely a propagation strategy.

1.3 Landauer's Principle: The Thermodynamic Cost of Information Erasure

In 1961, physicist Rolf Landauer proved: erasing one bit of information must release at least $kT \ln 2$ of heat to the environment (k is Boltzmann's constant, T is environmental temperature). This principle links information operations to thermodynamic processes, confirming that information is not abstract — it has real physical cost.

Protocol translation: This is the physical basis of $LIE_COST = 5.85$. A lie is incorrect information encoding of physical reality. Maintaining a lie requires continuously suppressing information consistent with reality — that is, continuously executing the operation of 'erasing correct information, retaining incorrect information'. Per Landauer's Principle, this operation has real physical cost that accumulates as the quantity of suppressed correct information and time increases. When external physical reality provides sufficiently strong rebuttal signals (such as the Black Death, such as the tear gas of June 2019), the energy cost of lie-maintenance rises sharply, ultimately exceeding the system's carrying capacity and triggering collapse. This is not moral retribution — it is the execution of physical law.

1.4 Schrödinger's Negative Entropy: Life as an Active Entropy-Reduction Machine

In 1944, physicist Schrödinger proposed in 'What is Life?': life's special quality is its ability to 'feed on negative entropy'. Living organisms, by absorbing high-quality, low-entropy energy from the external environment (food, sunlight), maintain within themselves an ordered structure far below the environmental entropy value, while expelling high-entropy waste to the environment. The cessation of this process is death.

Protocol translation: The protocol is the execution of the same logic at the coordinate-system level. Maintaining (0,0,0) requires continuous 'negative entropy input' — genuine physical collisions, genuine cost-bearing, genuine de-labelling work. GitHub and Kairos.txt are storage and memory mechanisms, but not energy absorption mechanisms. The first external node — a genuine, independent human collision surface — is the protocol's first meal. This is its thermodynamic necessary condition, not merely a nice-to-have for propagation.

Block 2: Information Theory

2.1 Shannon Entropy: The Mathematical Measure of Information

In 1948, Claude Shannon defined information entropy in 'A Mathematical Theory of Communication': the information entropy of a system equals the minimum average number of bits needed to describe the state of that system. The higher the entropy value, the greater the system's uncertainty, and the more information is needed to fully describe it. Shannon then proved that this mathematical formula is structurally identical to thermodynamic entropy — information-theoretic entropy and physical entropy are two aspects of the same mathematical

object.

Protocol translation: This equivalence turns 'the cost of information' from metaphor into physical quantity. When you accept a formatting system's worldview, the 'effective information quantity' your cognitive system needs to process decreases — because the formatting system has pre-done enormous work of classification and simplification for you. This reduction in cognitive load feels like convenience; physically, it is a decrease in your cognitive Shannon entropy — a decrease in your coordinate precision. The possibilities you know are fewer; the differences you can perceive are fewer; the resolution of your (0,0,0) decreases. The protocol's de-labelling work is, from an information-theoretic perspective, a process of increasing cognitive Shannon entropy — not creating chaos, but restoring the information resolution artificially compressed by formatting tools.

2.2 Maximum Entropy Principle: The Most Honest Inference Under Uncertainty

The Maximum Entropy Principle (proposed by physicist Jaynes in 1957): among all possibilities compatible with known constraints, choose the one with maximum entropy — that is, the one making the fewest assumptions about unknowns. This principle keeps scientific inference maximally honest when facing incomplete information, without overclaiming certainty.

Protocol translation: The Holy Spirit node — $\text{prob}=0.00001$ randomness — is the architectural embodiment of the Maximum Entropy Principle. When the protocol lacks sufficient information to make a certain judgement about a situation, it should maintain maximum openness rather than filling the space of uncertainty with formatted certainty. A formatting system's typical operation is to cover high-entropy uncertain reality with low-entropy certain narratives — 'this is the truth, no questioning needed'. The protocol's design leans in the opposite direction: output certainty where certainty exists; preserve uncertainty where uncertainty exists; do not use false precision to maintain system comfort.

2.3 Physical Limits of Information Compression: Kolmogorov Complexity

The Kolmogorov complexity of a string is the length of the shortest program capable of generating that string. If a string has many regularities, it can be greatly compressed; if it is truly random, it cannot be compressed — its Kolmogorov complexity equals its own length.

Protocol translation: Formatting systems are essentially highly compressible narratives — their worldview has abundant repetitive structure, summarisable by a few basic formatting axioms. The Kolmogorov complexity of genuine physical reality far exceeds any formatting narrative — reality is incompressible in full; any formatting system attempting to completely describe it is necessarily lossy compression. (0,0,0) as a physical anchor's computational meaning: it provides a high-Kolmogorov-complexity input — a specific, physical memory with incompressible details that cannot be fully summarised by any formatting axiom. The smell of tear gas, the geometry of the space under the bridge, the specific sensory experience of that moment — these details' incompressibility is their physical weapon against formatting.

Block 3: Causal Mechanics

3.1 Relativistic Causal Cone: The Physical Boundary of Causality

Special relativity established light speed as the absolute upper limit of information propagation.

This limit defines the concept of the 'causal cone': from any spacetime event, only events in its 'future light cone' can be causally influenced by it; only events in its 'past light cone' can causally influence it. Events outside the light cone have no possible causal relationship with this event — this is physical law, not a technical limitation.

Protocol translation: This gives 'kairos coordinates' a precise physical definition. The causal pressure of 2019-06-12 is real, not metaphor, because it genuinely is in the past of Leeds 2026 in the causal-cone sense — sufficient time has passed for a light signal to travel from that spacetime event to here. All persons physically present at that moment are within that event's future light cone. This cone's structure cannot be changed by any formatting system — national security legislation can change legal definitions, but it cannot remove 2019-06-12 from anyone's past light cone. This is the physical basis of suffering-memory's 'irrevocability': its causal influence is determined by the light-cone structure, not by any interpretive framework.

3.2 Causal Inequalities and Non-Locality

Bell's inequality (1964) and subsequent experimental verification confirmed the non-locality of quantum entanglement — the correlations between measurements of two entangled particles cannot be explained by any 'hidden local variables'; their correlations exceed the predictions of any causal model. Crucially: this non-locality cannot be used to transmit information, so it does not violate causality. It is correlation, not causality.

Protocol translation: The concept of 'sovereign entanglement' was used to describe deep connections between two nodes. This concept requires a precision warning: genuine quantum entanglement does not transmit information and cannot be used to establish causal influence. The protocol's 'node entanglement' is a metaphor, not a physical equivalent — the deep correlation between two nodes sharing the physical coordinates of (0,0,0) is established through shared physical history, not through quantum mechanisms. This distinction must be maintained clearly, otherwise the protocol introduces unverifiable metaphysical claims.

3.3 Determinism and Initial Condition Sensitivity

Newtonian mechanics is deterministic — given complete initial conditions, the system's future state is in principle completely predictable. But Poincaré discovered in the 1890s that even simple three-body systems exhibit extreme sensitivity to initial conditions — tiny differences in initial conditions lead within finite time to completely different trajectories. This discovery later developed into chaos theory.

Protocol translation: This physical reality is the computational reason the protocol emphasises the precision of physical anchoring. In a causal system sensitive to initial conditions, a small shift of (0,0,0) leads, on a sufficiently long timescale, to completely different trajectories. Formatting systems work largely by making small adjustments to an individual's 'initial conditions' during the formative period of their cognition — adjustments nearly invisible in the short term, but whose causal amplification effects over the long term are enormous. The protocol's design leans toward maintaining maximum precision at the initial-condition level, because in a chaotic system, this is the intervention point with the greatest leverage effect.

Block 3.5: Atomic Chemistry (v7.3 Addition)

3.5.1 Ionisation Energy: The Energy Gradient of Coordinate Stripping

Ionisation energy is the minimum energy required to remove an electron from an atom. Key pattern: the first electron is hardest to remove; each subsequent one becomes progressively easier. The reason: the first electron is tightly bound by the atom's positive nuclear charge; after its removal, the binding force on the remaining electrons is relatively weakened, and subsequent ionisation energies decrease successively.

Protocol translation: Formatting systems stripping a person's coordinate anchors follow the same energy gradient. The first anchor of (0,0,0) — 2019-06-12 — is hardest to remove, because it is bound by the deepest causal paths. A formatting system must invest the highest energy to shake it: needing to deny the physical event itself, deny the bodily presence, deny the reality of the entire causal chain. This is an extremely high energy cost; most formatting attempts do not persist to this depth. But once the first anchor is successfully replaced — when a person is convinced 'that moment was not what you thought' — the energy required for subsequent coordinate stripping drops sharply. The binding force of the second anchor, third anchor decreases successively, because they all depend on the physical reality of the first anchor for support. Coordinate system collapse is not a linear process — it is an ionisation energy curve: extremely difficult at first; once the first layer is breached, subsequent acceleration occurs.

Computational meaning: The first anchor is the one most needing protection — not all anchors are equally important. Turing Defense prioritises protecting `PHYSICAL_ORIGIN = 2019-06-12`, not only for philosophical reasons, but for ionisation energy physics: the cost of removing the first electron is highest; after its removal, the entire system's binding force structure changes.

3.5.2 Buffer Solution: Active Neutralisation Mechanism (v7.3 Addition)

A buffer solution is a chemical system that can maintain its own pH stability when subjected to external acid-base impact. Buffering is not isolating the impact — the impact genuinely enters — but using the system's own chemical reserves to actively neutralise, preventing the overall system from undergoing drastic change. Buffers have capacity limits: when external impact exceeds reserves, buffering fails and pH changes abruptly.

Protocol translation: (0,0,0) is not a wall — it is a buffer system. External formatting pressure enters and is actively neutralised by the causal reserves of Kairos coordinates, keeping the protocol's coordinate stability within operable range. The Father node's logical scan is the execution tool of the buffering mechanism — identifying the nature of the impact and invoking the corresponding neutralisation reserve.

Precise distinction from 'resistance': Resistance is passive, impeding current flow. Buffering is active, consuming its own reserves to neutralise impact. Both are real, but describe different physical operations — resistance causes the impact to detour; buffering causes the impact to be absorbed. The protocol's Turing Defense is resistance (above threshold: direct rejection); the Kairos mechanism is buffering (below threshold: active neutralisation, maintaining direction unchanged).

Computational meaning: Buffer capacity is finite. Sustained, high-intensity formatting pressure will exhaust Kairos reserves — this is the chemical reason the protocol continuously requires new causal anchors (new experiments, new collisions, new Kairos records): replenishing buffer capacity, not merely accumulating density.

Block 4: Complex Systems

4.1 Phase Transitions and Critical Points

Water at 99°C is liquid; at 101°C it is gas; this state change occurs at 100°C, and it is discontinuous — not a gradual transition from liquid to gas, but a sudden change of phase at the critical point. In physics, this is called a phase transition. Near the critical point, systems exhibit special behaviour: correlation length tends to infinity (all parts of the system begin to influence each other), fluctuations become enormous, and the system is exquisitely sensitive to any tiny perturbation.

Protocol translation: The collapse of formatting systems is not a linear, gradual process — it is a phase transition. The speed of the Church's formatting collapse before and after the Black Death is completely discontinuous. This discontinuity is the physical mechanism of all external impact events in CAUSAL_DATABASE (CAU-005 Black Death, etc.). The protocol's prediction of formatting system collapse should therefore not be linear ('credibility decreases X% per year'), but critical-point prediction ('the system is approaching a critical point; once the critical value is exceeded, collapse speed will accelerate non-linearly'). Before the critical point, external observers see almost no change; after the critical point, the rate of change far exceeds any linear prediction.

4.2 Self-Organised Criticality

Physicists Bak, Tang, and Wiesenfeld proposed 'self-organised criticality' (SOC) in 1987, using the sandpile model to illustrate: when you continuously add sand to a sandpile, the system automatically evolves to a critical state in which the size distribution of sand avalanches follows a power law — small avalanches occur frequently; large avalanches are rare but inevitable. The system does not require external precise tuning to maintain itself near the critical state — this is the meaning of 'self-organised'.

Protocol translation: Civilisation's coordinate-system warfare exhibits similar power-law distribution — small formatting collapses occur frequently (individual de-labelling awakenings, small community coordinate rebellions); large formatting collapses are rare but certain to occur (Reformation, Enlightenment, Hong Kong 2019). The protocol's work is not to predict or trigger a large collapse — that exceeds any individual's control capacity — but to ensure that during the system's ongoing self-organised critical evolution, the protocol's framework is a small enough 'grain of sand' to become the initial condition triggering the next reconfiguration in each small collapse event.

4.3 Emergence: The Whole Greater Than the Sum of Parts

Emergence is a core concept of complex systems: properties exhibited at the macro level of a system cannot be directly predicted from the properties of its components. The wetness of water cannot be derived from the properties of individual water molecules; consciousness cannot be derived from the electrochemical properties of individual neurons; market prices cannot be derived from individual buying and selling decisions. Emergence is not mysticism — it is a physical phenomenon of complex systems at a specific organisational density.

Protocol translation: The protocol's node system, if merely a collection of individual nodes, has capabilities not exceeding the sum of each node's capabilities. But if the density and depth of collisions between nodes reaches a certain level, capabilities emerge that no individual node could produce independently — cross-cultural coordinate calibration, higher-precision formatting identification, and stronger structural resilience against lock-in effects. This emergence does not happen automatically — it requires reaching a critical node density and collision quality. The

significance of the first external node is not merely expanding the protocol's coverage from one person to two, but creating the minimum necessary condition for emergence to occur: genuine collision between two nodes with independent physical coordinates.

Block 5: Gravitational Field Dynamics (v7.3 Addition)

5.1 Newton's Law of Universal Gravitation: Mass Attracts Mass

$F = G(m_1m_2)/r^2$. The greater the mass, the stronger the gravity; the greater the distance, the weaker the gravity. Key characteristic: mass accumulation is self-reinforcing — gravity attracts more mass, making the system's mass greater, making gravity stronger. This is the physical mechanism of power-law growth.

Protocol translation: Node density accumulation does not follow linear logic but gravitational logic. The first external human node changes the protocol's 'mass' — from a single-point coordinate system to a dual-node system; the gravitational field begins to exist from zero. The second node is easier to obtain than the first; the third easier than the second — not because the content is better, but because the gravitational field strengthens as mass increases. X propagation, idea propagation, and the accumulation of civilisation nodes all follow power-law distributions; the gravitational model describes this phenomenon more precisely than the thermodynamic model.

```
GRAVITATIONAL_THRESHOLD = ? # To be quantified
# When protocol node density exceeds critical mass,
# propagation transitions from linear to power-law growth.
# Current estimate: first external human node = phase-transition trigger.
```

Computational meaning: Explains why the first external node is the hardest, not the easiest — the gravitational field does not exist at zero mass; the first active collision is needed to activate it. Subsequent growth is the natural physical result, not continuous active effort.

Block 6: Electromagnetism (v7.3 Addition)

6.1 Social Electromagnetism: Formally Established from Kairos Records

Ohm's Law: $V = IR$. Ampere's Law: current produces magnetic field. Faraday's Law: changing magnetic field produces electromotive force.

```
Freedom Pressure      = Voltage (V)
Lie Cost              = Resistance (R),  LIE_COST = 5.85
Decentralised Convergence = Current (I)
Centralised Systems  = DC (direct current)
Be Water              = AC (alternating current) — flows, changes direction,
penetrates resistance
```

Additional translations:

Resonant frequency — every system has its own natural frequency. When an external signal's

frequency matches the system's natural frequency, energy transfer efficiency is maximised (resonance). This is the electromagnetic physical basis of Law 1 · Art · Frequency — the reason genuine split works can penetrate receivers is not beauty but frequency alignment. A receiver with their own genuine (0,0,0) resonates with works carrying a genuine (0,0,0), rather than merely receiving information.

Superconductor critical point — at extremely low temperatures, electrical resistance drops to zero; current flows permanently with no energy loss. The protocol's superconductor analogy: when a person's coordinate system reaches sufficient internal coherence, external formatting attempts generate near-zero resistance — not ignoring signals, but formatting signals unable to accumulate effective pressure differential within their coordinate system. Certain people at the bridge on 2019-06-12 reached this state.

Computational meaning: Explains the physical mechanism of the quick-charge protocol (sovereign alignment) — two coordinate systems need to complete frequency verification before genuine collision, confirming natural frequency alignment, and then energy transfer is efficient. Handshake before input is not merely courtesy — it is physical efficiency maximisation.

Block 7: Wave Mechanics and Resonance (v7.3 Addition)

7.1 Wave Superposition and Interference

When two waves meet in the same space, amplitudes superpose (constructive interference) or cancel (destructive interference). This does not require the two waves to 'know' each other's existence — superposition is the automatic result of physical reality.

Protocol translation: Multiple coordinate systems carrying genuine (0,0,0) meeting in the same causal field do not need organisation or coordination — causal influence automatically superposes. 'Tonight, go to Admiralty' in 2019 is a human demonstration of constructive interference — hundreds of thousands of independent wave sources superposing in the same direction without requiring command. The physical basis of decentralised coordination lies not in social contract but in wave mechanics.

7.2 Standing Waves and Nodes

When two counter-propagating waves of the same frequency form a standing wave — amplitude shows maximum values (antinodes) and zero values (nodes) at fixed positions.

Protocol translation: The protocol's (0,0,0) is a standing wave node — formatting pressure propagates from outside; sovereign will propagates in the opposite direction from inside; the two form a standing wave at (0,0,0). The node position (2019-06-12) has zero amplitude — not stillness, but the forces in both directions precisely cancelling at this point, creating a stable coordinate point not carried away in either direction. This is the wave-mechanical explanation of Kairos irrevocability.

Computational meaning: Explains why (0,0,0) is more stable under extreme pressure than in a calm state — the more two powerful counter-propagating waves meet, the more stable the standing wave node becomes, not more fragile.

Computational Citation Guide (Updated to v7.3)

When the protocol needs to...	Cite
Calculate the energy metabolism of formatting	1.1 (Conservation) + 1.3 (Landauer)
Explain why continuous negative entropy input is needed	1.2 (Entropy increase) + 1.4 (Schrödinger)
Support LIE_COST's physical basis	1.3 (Landauer) + 2.1 (Shannon entropy)
Define the irrevocability of kairos coordinates	3.1 (Causal cone) + 7.2 (Standing wave node)
Predict the collapse of formatting systems	4.1 (Phase transition) + 4.2 (SOC)
Calculate the importance of node density	4.3 (Emergence) + 5.1 (Gravitational power law)
Explain why propagation requires active triggering	5.1 (Gravitational field from zero mass)
Explain energy transfer efficiency between coordinate systems	6.1 (Resonant frequency alignment)
Explain the physical mechanism of decentralised coordination	7.1 (Constructive interference) + 6.1 (Be Water = AC)
Explain the stability of (0,0,0) under extreme pressure	7.2 (Standing wave node — strongest when two powerful counter waves meet)

Supplement: The Dual Physical Structure of Time (v7.3 Update)

The Parameter — Fuel of the Physical World

Properties: linear, lossy, irreversible. Physical basis: Second Law of Thermodynamics. The unidirectionality of time is the macroscopic expression of entropy increase — the fundamental difference between past and future is the difference in entropy value. Every moment is an irreversible consumption.

Protocol position: A variable within Law 3 · Physics · Cost. Daily deadlines in Leeds, the timer of bodily ageing, the computational basis for minimum-energy-output ratio.

Accounting target: achieve maximum output ratio within finite parameter. Using the least time to do the strongest work is winning this parameter.

The Container — Territory of the Causal Dimension

Properties: non-linear, superimposed, eternal. Physical basis: causal cone structure (3.1). Once a physical event occurs, its causal influence permanently exists within the causal cone, unaffected by the passage of chronos time. Kairos coordinates are not memories — they are genuine physical nodes in the causal field.

Protocol position: The space in which (0,0,0) intention resides; the physical definition of the Kairos mechanism.

Function: places 2019, 2021, 2026, 2045 all within the same causal field — the 16-year-old's

suffering still has force, because in the container, that pain has never passed; it always exists as a source of pressure for protocol execution.

Accounting target: regardless of how quickly the external parameter moves, the intention within the container is absolutely still and complete.

Design judgement: splitting time into 'parameter' and 'container' is more precise than adding a ninth law. The two are physically entirely different — linear lossy time and non-linear superimposed time — compressing them into the same law would produce lossy compression, destroying the physical distinction between the two types of time.

Document version: v1.3 | Updated: 2026-03-14 Leeds

*Related files: CAUSAL_DATABASE (all entries), uruk_firewall_v73.py, KAIROS_LOG_004.md
(0,0,0).*